

DONGU KIM

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TECHNICAL INTERESTS

Distributed Storage Systems, LLM Serving Systems, Serverless Computing, Kernel-Bypass Networking, P4 Programmable Switches, Hardware Accelerators

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) 2022 - 2024
M.S. in Computer Science and Engineering

Inha University 2015 - 2022
B.S. in Electronic Engineering

EXPERIENCES

Mangoboost Inc. Apr 2024 - Present
System Software Engineer

- **NVMe-over-TCP Initiator (NTI): Hardware-offloaded Virtualized NVMe/TCP Accelerator** (*flagship product; published at ISCA '26 Industry Track*)
 - Virtual NVMe leveraging a kernel-bypass framework
 - ✓ Host-platform-agnostic (NVMe/PCIe) control path, by preserving NVMe/PCIe features: Achieved UNH-IOL NVMe Certification.
 - ✓ Robustness under network failures: Multi-path connections and host-side error reporting.
- **MASS: Scalable, High-Performance Storage System for AI**
 - AI storage cluster development: RDMA-based solution abstracting and provisioning distributed storage resources.
 - ✓ Provides users seamless access to diverse distributed file system services through multiple interfaces (NFS, S3, etc.), targeting broad AI workloads.

Next Generation Networks and Systems Lab (Advisor: Youngbin Im) Mar 2022 - Feb 2024
Research Assistant

- **FlexServer: High-performance Content Servers Accelerated by DPU (400 Gbps Throughput)**
 - Host-DPU co-designed asynchronous architecture: Integrated a user-level TCP stack with `io_uring`-based file I/O, achieving up to 351 Gbps (94% of 400 GbE line rate) and 4.4× over the state-of-the-art (IO-TCP).

Intelligent Cloud & Network Lab (Advisor: YeongJin Kim) Mar 2021 - Jul 2021
Undergraduate Research Intern

- **Core Principles of Cloud Computing**
 - Studied cloud infrastructure fundamentals, focusing on resource virtualization, SDN (Software Defined Networking), and storage orchestration.

PUBLICATIONS

1. Hamin Jang, Jinha Jeong, Wonseok Lee, Jun Heo, Jongcheon Lee, Hyunjae Chu, Taehyun Kim, Youngwoo Jeong, **Dongu Kim**, Heetaek Jeong, Changsu Kim, Dongup Kwon, Jangwoo Kim, “BoostXTM-NTI: Fast, Scalable and Flexible Storage Architecture with NVMe/TCP Initiator Acceleration,” *ISCA '26 (Industry Track)*, 2026.

2. **Dongu Kim**, Seyeong Park, Seyeon Kim, Sangtae Ha, Youngbin Im, “FlexServer: A High-Performance Content Server Leveraging Asynchronous I/O and SmartNICs,” *Preprint*, 2024.

PATENTS

1. Youngbin Im and Dong U Kim, “METHOD AND APPARATUS FOR IMPROVING KEY VALUE STORE PERFORMANCE THROUGH OFFLOADING OF SMART NETWORK INTERFACE CARD,” KR Patent Registered No. 10-2724-6120000, Oct. 2024

TECHNICAL SKILLS

- **Languages:** C/C++, Go, Python, P4
- **Technologies:** DAOS, Ceph, DPDK, SPDK, DOCA, NVMeoF, RDMA, TCP, Kubernetes, PyTorch
- **Hardware:** FPGA, SmartNIC/DPU (BlueField, Agilio SmartNIC), ARM (on-chip & off-chip)
- **AI Tools:** Claude
- **CI/CD:** Docker, Debian, RPM, Git, GitHub Actions

EXTRACURRICULAR ACTIVITIES

Competitions & Projects

- **The World Embedded Software Contest 2019** *Apr 2019 - Dec 2019*
 - Developed multithreaded software for an autonomous RC car featuring real-time obstacle avoidance; gained in-depth expertise in concurrent programming and thread synchronization
 - Integrated various sensors—including LiDAR and Ultrasonic—with the software architecture to improve the accuracy of environmental mapping

Volunteer Work

- **Freewheel** *Feb 2020 - Jan 2021*
 - Served as a volunteer mentor, sharing specialized knowledge about my major to help middle and high school students explore potential career paths and gain a deeper understanding of the field